Name: _____

WORKSHEET

Rates of change – Gradients of secants

8	18	11	18	15	13	12	3	6	15	18	9	4							
16	11	14	15	6	12	11	5	9	17	15	13	6	16	15					
14	18	17	16	10	4	16	11	8	15	10	14	15	18						
9	4	15	6	16	15	19	4	10	7	4	2	15	1	14					

St Francis de Sales

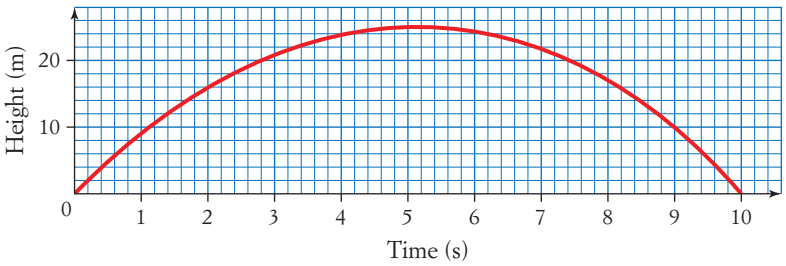
Rule

To calculate the average rate of change of a curved graph from x_1 to x_2, draw a secant that passes through both points. The gradient of the secant is equal to the average rate of change between the two points.

Average rate of change from x_1 to $x_2 = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$

Questions

A ball is catapulted into the air and its height at various times is shown in the graph below.



What is the average rate of change during the following times?

1 From $t = 0$ to $t = 2$

2 From $t = 4$ to $t = 6$

3 From $t = 5$ to $t = 7$

4 From $t = 6$ to $t = 10$

Solutions

A 16

B 2 cm/week

C 0 m/s

D 5 m/s

E −6 m/s

F 7 m/s

G 1 m/s

H 4 m/s

I 6 cm/week

L 8 m/s

N 4 cm/week

O 10

P 30

R 3 cm/week

S −2 m/s

T 6

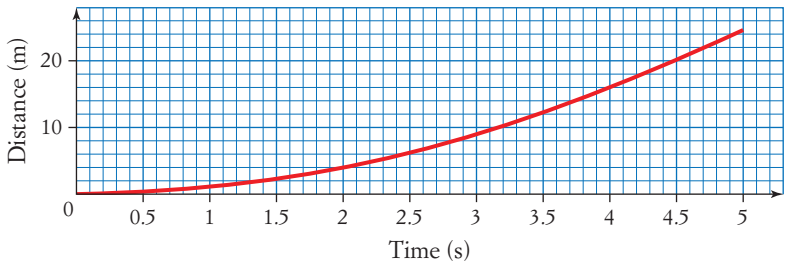
U 4

W 9

Y 15

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The distance travelled by a cyclist in the first 5 s after taking off is shown in the graph below.



What is the average rate of change during the following times?

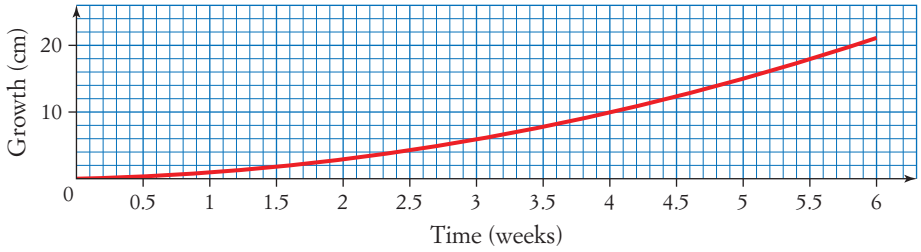
5 From $t = 0$ to $t = 1$

6 From $t = 1$ to $t = 3$

7 From $t = 2$ to $t = 5$

8 From $t = 0$ to $t = 5$

The growth of a tomato seedling is recorded in the graph below.



What is the average rate of change during the following times?

9 From $t = 0$ to $t = 3$

10 From $t = 1$ to $t = 4$

11 From $t = 2$ to $t = 5$

12 From $t = 5$ to $t = 6$

Find the average rate of change for the function $f(x) = x^2 + 3x - 1$:

13 from $x = 1$ to $x = 5$

14 from $x = 2$ to $x = 10$

15 from $x = 0$ to $x = 3$

16 from $x = 6$ to $x = 7$

Find the average rate of change for the function $g(x) = x^3 - 2x^2 + 5x - 3$:

17 from $x = 0$ to $x = 1$

18 from $x = 1$ to $x = 3$

19 from $x = 2$ to $x = 5$

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